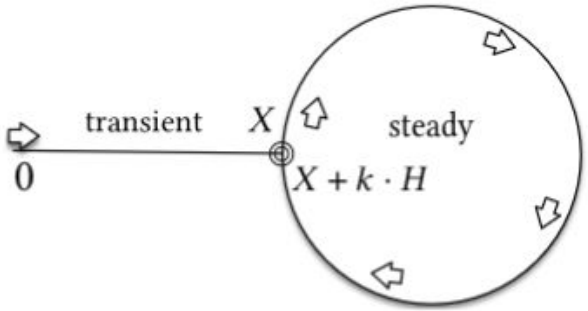
Project SHOMEN

how long should we run the simulation ?

Simulation Interval

- Finite time interval
- The schedule starts to repeat a cycle
- reach all possible states (reachable)



System Model

on a mono-processor system with DM (Deadline Monotonic) scheduler ,
we have these tasks :

Task	C (WCET)	Period / P	Deadline / D	Offset / O
T_1	1	4	3	0
T_2	1	5	4	2
T_3	2	10	8	4
T_4	4	10	9	3
T_5	1	10	7	7

System Model

Priority Ordering according to DM :

- T_1
- T_2
- T_5
- T_3
- T_4

Offset Max :

$O_{max} = 7$

Hyperperiod :

$H = LCM(T_1, T_2, T_3, T_4, T_5) = 20$

Application: Leung et al., 1982

System Assumptions Verification:

- **Processor Architecture: mono-processor** system.
- **Scheduling Policy: EDF** or **FTP (Fixed Task Priority)** scheduler => The system uses a **DM (Deadline Monotonic)**
- **Task Arrival (Asynchronous):** Tasks having varying initial offsets
- **Task Independence:** no shared resources or precedence constraints.
- **Deadline Constraints** $D_i \leq T_i$

Application: Leung et al., 1982

- **Formula**: $O_{max} + 2H$

Simulation Interval Upper Bound: **47**

Application: Goossens et al., 1999

System Assumptions Verification:

- **Processor Architecture: mono-processor** system.
- **Scheduling Policy: FTP (Fixed Task Priority)** scheduler => The system uses a **DM (Deadline Monotonic)**
- **Task Arrival (Asynchronous):** Tasks having varying initial offsets
- **Task Independence:** no shared resources or precedence constraints.
- **Deadline Constraints** $D_i \neq T_i$

Application: Goossens et al., 1999

- **Formula**: $O_{max} + 2H$

Simulation Interval Upper Bound: **47**

Application: Goossens et al., 1997

System Assumptions Verification:

- **Processor Architecture: mono-processor** system.
- **Scheduling Policy: FTP (Fixed Task Priority)** scheduler => The system uses a **DM (Deadline Monotonic)**
- **Task Arrival (Asynchronous):** Tasks having varying initial offsets
- **Task Independence:** no shared resources or precedence constraints.
- **Deadline Constraints** $D_i \leq T_i$

Application: Goossens et al., 1999

- **Formula**: $S_n + H$

S_n : Stabilization time

Assume that tasks are ordered by priority level

$$S_1 = O_1$$

$$S_i = max(O_i, O_i + \left\lceil \frac{S_{i-1} - O_i}{T_i} \right\rceil)$$

Application: Goossens et al., 1997

- $S_1 = O_1 = 0$
- $S_2 = \max(O_2, O_2 + \lceil (S_1 - O_2)/T_2 \rceil) = 2$
- $S_5 = \max(O_5, O_5 + \lceil (S_4 - O_5)/T_5 \rceil) = 7$
- $S_3 = \max(O_3, O_3 + \lceil (S_2 - O_i)/T_3 \rceil) = 5$
- $S_4 = \max(O_4, O_4 + \lceil (S_3 - O_4)/T_4 \rceil) = 4$

Simulation Interval Upper Bound: $S_5 + H = 24$

Application: Goossens et al., 2022

System Assumptions Verification:

- **Scheduling Policy:** Work-conserving deterministic and memory-less scheduler
- **Task Arrival (Asynchronous):** Tasks having varying initial offsets
- Existance of a preemption delay
- **Deadline Constraints** $D_i \neq T_i$

Simulation interval for general cases

$$\boldsymbol{H \cdot (n+1) \cdot (\alpha_{max}+1) \times \prod_{i=1}^{n} ((O_i + D_i - T_i)_0 + 1)}$$

α_{max} : the preemption cost

The authors also state that :

- For EDF, the Simulation interval UB = $\mathrm{O_{max}} + 2\mathrm{H}$
- For FTP scheduler, the Simulation interval UB = $S_n + H$

Application: Goossens et al., 2022

- DM => FIFO
- Table of preemption delay for each task :

Task	α / preemption delay
T_1	1
T_2	1
T_3	1
T_4	2
T_5	2

$$H \,.\, (n+1) \,.\, (\alpha_{max}+1) \,.\, \prod ((O_i + D_i - T_i)_0 + 1)$$

$$= 20 \times 6 \times 3 \times (1 \times 2 \times 3 \times 3 \times 1)$$

$$= 6480$$

Simulation Interval Upper Bound: 6480

Choquet-Geniet et al., 2004

System Assumptions Verification:

- **Scheduling Policy:** mono-processor , Work-conserving scheduler
- **resource-sharing tasks with precedence constraints**
- **Task Arrival (Asynchronous)**
- **Deadline Constraints** $D_i \leq T_i$

Simulation interval UB : $\boldsymbol{\theta} + \boldsymbol{H} \leq O_{max} + 2H$

θ: latest acyclic idle time

$$\theta \leq O_{max} + H$$

Before let's introduce these functions :

- *Release(t)* is the sum of the computing times of the tasks that are released at time *t*
- *Request(t)* is the sum of the remaining computing times of the pending tasks at time *t*

Algorithm of computation of Θ

```
Req:= 0;
idle_slots:= empty queue; -- contains the dates of the idle
slots occurring within the interval [tfirst, t]
nb_i:= 0; -- number of idle slots within the considered
interval, i.e. length of idle slots
tfirst := 0; -- begining of the considered interval
t:= 0; -- end of the considered interval
last := -1; --date of the last known acyclic idle slot
found := false;

While (t < r + P) and not found DO
  Req:= Req + Release(t);
  IF Req = 0
    THEN -- There is an idle slot
      nb_i:= nb_i + 1;
      queue (idle_slots, t);
    ELSE Req:= Req - 1; -- the processor runs for one time unit
  END IF;
  IF t - tfirst + 1 = P -- the considered interval is of size P
  THEN
    IF nb_i > P(1 - U) -- some of the idle slots are acyclic
    THEN
      last := idle_slots[nb_i - P(1 - U)] -- the first idle
      slots of the interval are acyclic
    ELSE -- the interval contains no acyclic idle slots. The last
      acyclic idle slot has already been found
      found := true;
    END IF;
    tfirst := tfirst + 1; -- the begining of the interval is moved
                          -- of one time unit
    IF head(idle_slots) = tfirst - 1 -- the first idle slot in
      the queue does no more belong to the considered interval, thus
      it must be removed from the queue
    THEN
      unqueue(idle_slots);
      nb_i := nb_i - 1;
    END IF;
  END IF;
  t := t + 1; -- the end of the interval is moved
              -- of one time unit
END DO;
t_c:= last;
r_0:= last + 1;
```

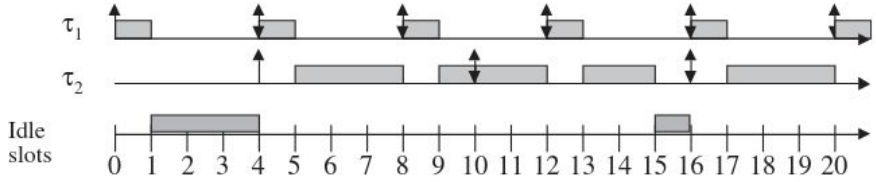
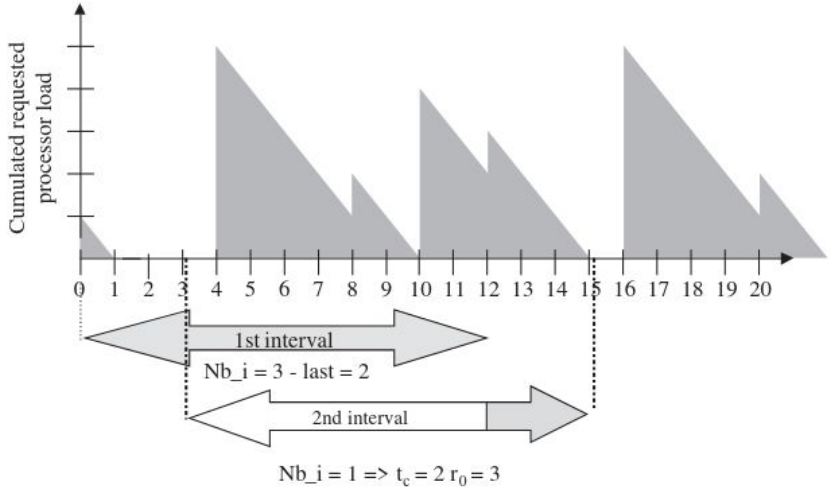


Fig. 4. RM schedule for the system $S_2 = \{\tau_1(0, 1, 4, 4), \tau_2(4, 4, 6, 6)\}$.



Example given by the authors

Application: Guidolin-Pina et al., 2026

System Assumptions Verification:

- **Scheduling Policy:** Work-conserving deterministic and memory-less scheduler
- **Task Arrival (Asynchronous):** Tasks having varying initial offsets
- **Deadline Constraints** $D_i \leq T_i$

Simulation interval UB: $\boldsymbol{t_0 + H(1 + U)} \leq O_{max} + 2H$

t_0: is the instant just before the set of clocks behave cyclically

$$t_0 = \max(0^-, \max(\mathrm{O_i} - \mathrm{T_i}))$$

Note: only need to simulate up to the first idle point strictly after $t_0 + H$

Application: Guidolin-Pina et al., 2026

$t0 = \max(0^-, \max(O_i - T_i)) = \max(0^-, -3) = 0^-$
$U = (1/4) + (1/5) + (2/10) + (4/10) + (1/10) = 1.15$

Simulation Interval Upper Bound: $0 + 20(\ 1 + 1.15) =$ **43**